Towards Capability-Aware Traversability Navigation for Unstructured Environments

Gianluca Capezzuto, Felipe Tommaselli, Matheus P. Angarola, Ricardo V. Godoy, Marcelo Becker
University of São Paulo, Brazil

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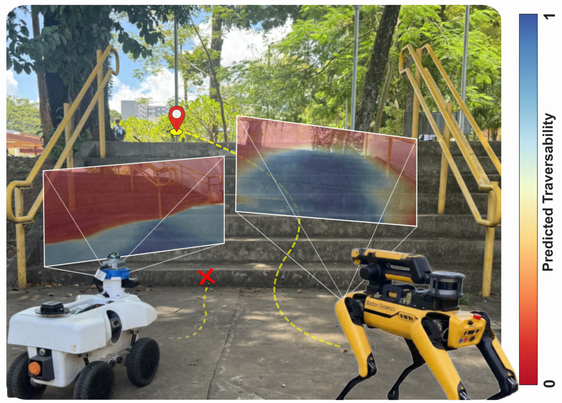
1 Problem & Motivation

Traversability depends on the robot's capabilities.



This requires solving two coupled problems:

Represent robot capabilities

Ground terrain semantics in **robot capabilities**

⟷

Predict robot traversability

Predict dense **traversability** for the target robot

Previous works follow two routes:

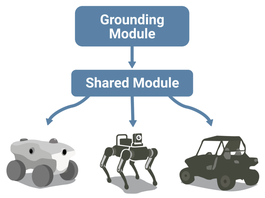
1 Capability implicit in demonstrations

Demonstrations
(robot behavior)

↓

Learned Traversability Predictor

2 Per-embodiment learned grounding

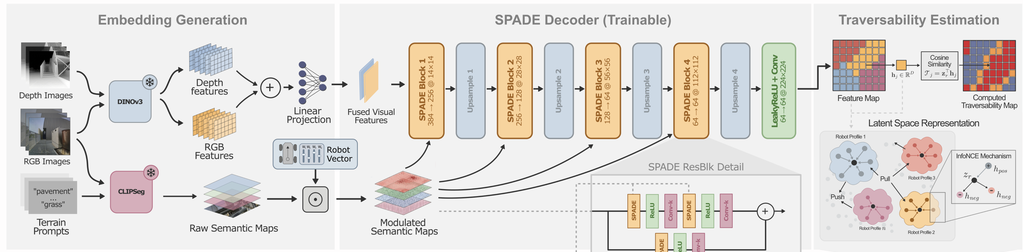


One shared grounding network, then distribute to each robot

? What if one **shared traversability predictor** could adapt to **different robot capabilities** without training a **separate predictor** or **grounding module** for each robot?

2 Method: Capability-Aware Traversability (CAT)

A shared predictor is modulated by a robot capability representation to estimate robot-specific traversability.



Visual-semantic encoding

DINOv3 encodes RGB-D images. CLIPSeg grounds terrain semantics.

Capability conditioning

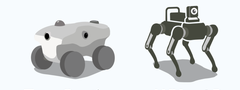
Robot capabilities modulate semantic features for robot-specific traversability.

SPADE decoding & readout

Conditioned semantics guide dense prediction against a robot-specific prototype.

Trajectory-grounded supervision

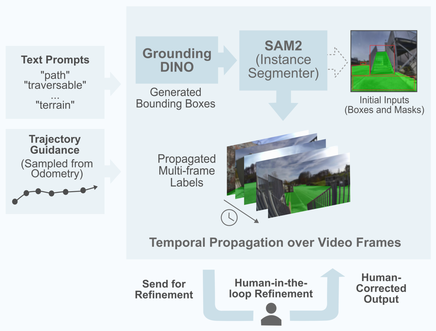
1 Multi-Source Data Acquisition



TerraSentia (Small Wheeled) **ANYmal D** (Quadruped)

ATV (Off-road Car)

2 Grounded SAM 2 Label Engine

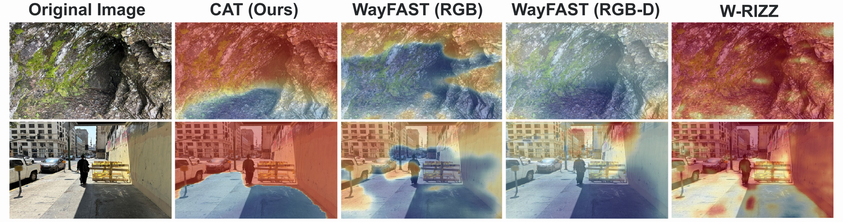


3 Dense Training Masks

Human-refined dense masks

Executed trajectories anchor dense masks, expanded with GroundingDINO, SAM 2, temporal propagation, and human refinement.

3 Main Results



+11.0% AUROC vs. strongest baseline

+15.8% AUPRC on human traces

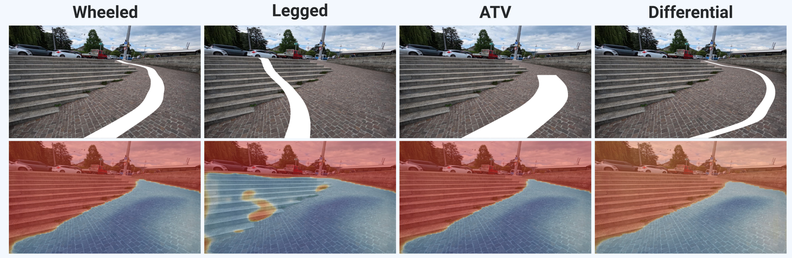
+14.2% Traversability with capability conditioning

Key results for CAT

Evaluation setting	AUROC	AUPRC
NaviTrace	0.794	0.169
Held-out trajectories	0.945	0.832

Best overall path alignment across both evaluation settings.

Qualitative Results



The same scene yields **different traversability predictions depending on the robot's capabilities**, showing the effect of capability conditioning.

Ablation

Capability conditioning matters

Method	Emb-W	Emb-L	AUROC
Uniform capability	0.543	0.517	0.950
Concatenation	0.541	0.524	0.944
Global Modulation	0.549	0.530	0.951
CAT (Full)	**0.567**	**0.540**	**0.945**

Full CAT achieves the best embodiment-awareness scores.

Robot-specific prototypes matter

Method	Emb-W	Emb-L	AUROC
Shared prototype	0.540	0.497	0.942
Multiple prototypes	0.550	0.521	0.938
CAT (Full)	**0.567**	**0.540**	**0.945**

Profile-specific prototypes outperform shared or multiple-prototype alternatives.

One shared predictor, different robot capabilities, different traversability maps.

Towards more **capable robots** in the real world.

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